

NETWORK VIRTUALIZATION

An Introduction

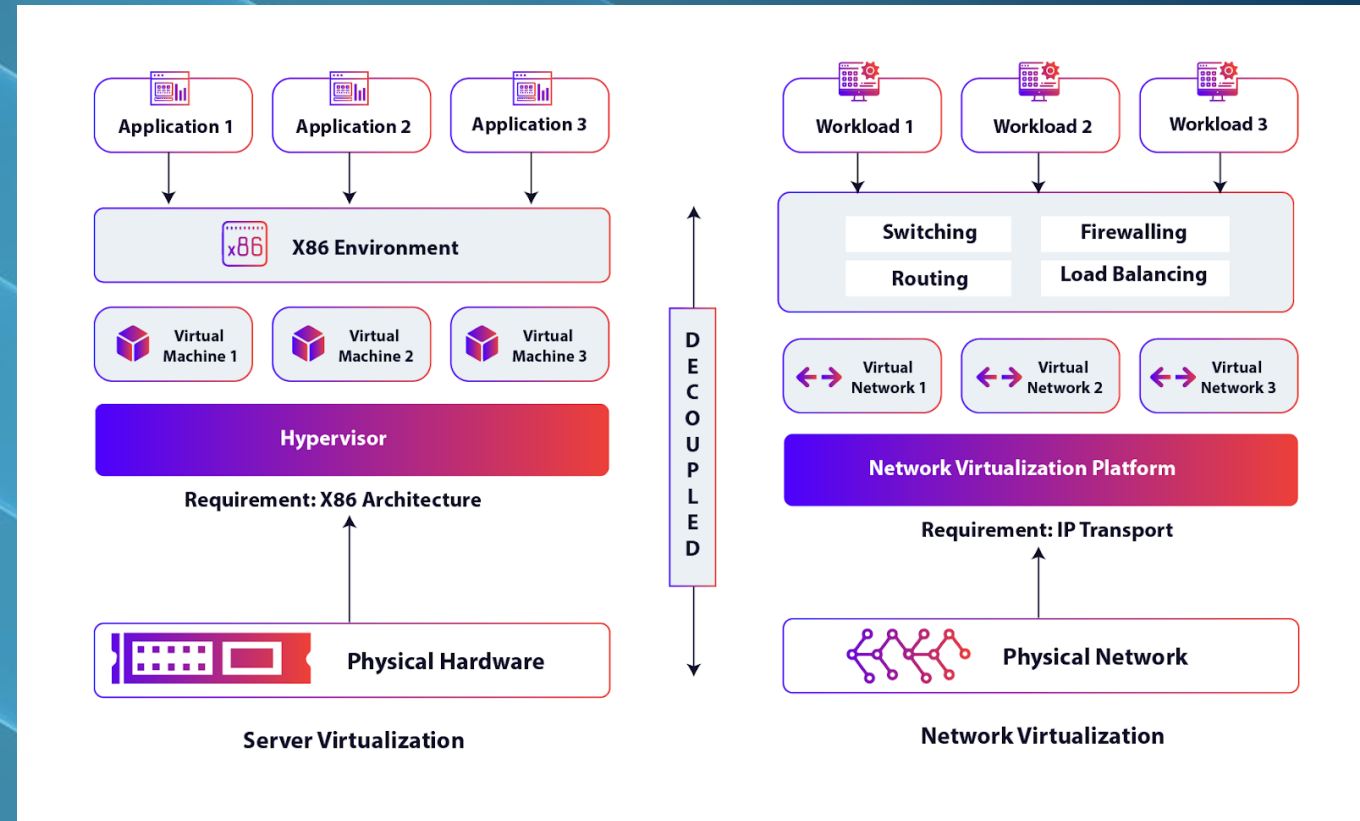


- ▶ Definition
- ▶ Types
- ▶ Architecture
- ▶ Benefits & Challenges

AGENDA



- Abstracting network resources that were traditionally delivered in hardware to software
- Combine multiple physical networks to one virtual, software-based network
- Can divide one physical network into separate, independent virtual networks
- To move virtual machines across different domains without reconfiguring the network

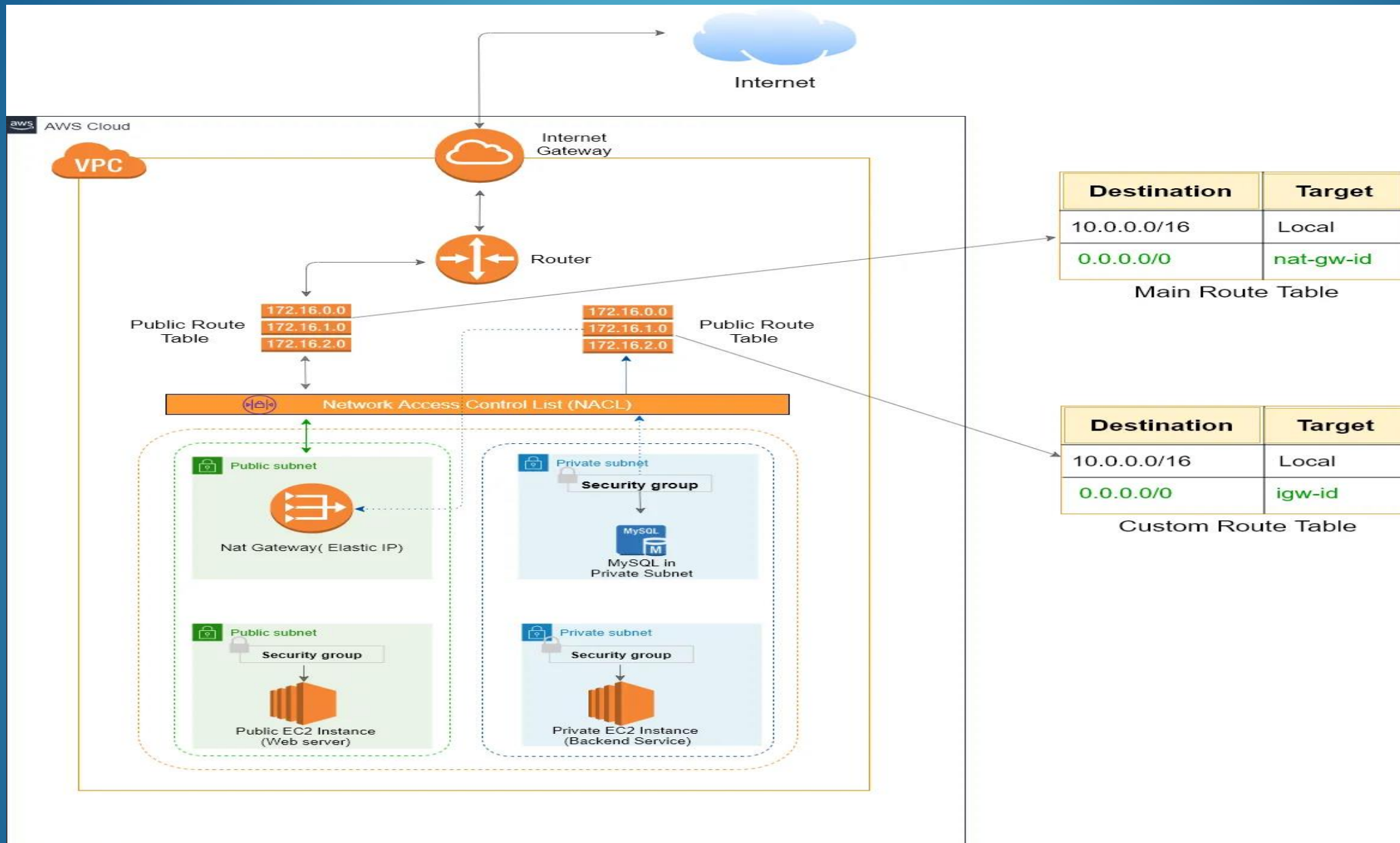


DEFINITION

- External network virtualization
 - combine systems physically attached to the same LAN
 - conversely divide separate LANs into the same
- Internal network virtualization
 - acts within a server to emulate a physical network
 - to improve a server's efficiency
 - involves configuring a server with software containers

TYPES





VPC ARCHITECTURE

Benefits

- Reduces cost
- Enhances resilience
- Reduces downtime
- Centralized control
- Increases productivity and efficiency
- Network visibility

Challenges

- New skills for IT staff
- Lower performance
- Changing standards
- Difficult management
- Finding the right vendor

BENEFITS & CHALLENGES



DEMONSTRATION

Launching and accessing a virtual network in AWS

HANDS-ON EXERCISE

- Create a VPC in AWS
- Launch a VM in within VPC
- Install 'docker' in VM
- Install a NGINX web service in VM
- Access from outside of VM the web service



THANK YOU !

